
Review Exercise

2. Define the following: (i) relative frequency (ii) expected frequency

Relative Frequency: Relative frequency shows how often a specific event happens compared to the total number of events or trials.

It is calculated using this formula:

$$\begin{aligned}\text{Relative Frequency} &= \frac{\text{Frequency of specific event}}{\text{Total frequency}} \\ &= \frac{x}{N}\end{aligned}$$

Where:

- x = frequency of the specific event
- $N = \sum f$ = total of all frequencies

Expected Frequency: Expected frequency is a measure that estimates how often an event should occur, based on its probability.

Expected frequency is calculated using the formula:

$$\begin{aligned}\text{Expected frequency} &= \text{Total number of trials} \times \text{Probability of the event} \\ E &= N \times P(A)\end{aligned}$$

3. An urn contains 10 red balls, 5 green balls and 8 blue balls. Find the probability of selecting at random. (i) a green ball (ii) a red ball (iii) a blue ball (iv) not a red ball (v) not a green ball

The total number of balls in the urn

Muhammad Tayyab (Govt Christian Daska)

$$\begin{aligned}\text{Total balls} &= \text{Red} + \text{Green} + \text{Blue} \\ n(S) &= 10 + 5 + 8 \\ n(S) &= 23\end{aligned}$$

(i) Let A be the event of selecting a green ball.

$$\begin{aligned}A &= \{\text{green balls}\} \\ n(A) &= 5 \\ P(A) &= \frac{n(A)}{n(S)} \\ P(A) &= \frac{5}{23}\end{aligned}$$

(ii) Let B be the event of selecting a red ball.

$$\begin{aligned}B &= \{\text{red balls}\} \\ n(B) &= 10 \\ P(B) &= \frac{n(B)}{n(S)} \\ P(B) &= \frac{10}{23}\end{aligned}$$

(iii) Let C be the event of selecting a blue ball.

$$\begin{aligned}C &= \{\text{blue balls}\} \\ n(C) &= 8 \\ P(C) &= \frac{n(C)}{n(S)} \\ P(C) &= \frac{8}{23}\end{aligned}$$

(iv) Let D be the event of selecting a red ball.

$$\begin{aligned}D &= \{\text{red balls}\} \\n(D) &= 10 \\P(D) &= \frac{n(D)}{n(S)} \\P(D) &= \frac{10}{23}\end{aligned}$$

To find probability not selecting a red ball, we have

$$\begin{aligned}P(D') &= 1 - P(D) \\P(D') &= 1 - \frac{10}{23} \\P(D') &= \frac{23 - 10}{23} \\P(D') &= \frac{13}{23}\end{aligned}$$

(v) Let E be the event of selecting a green ball.

$$\begin{aligned}E &= \{\text{green ball}\} \\n(E) &= 5 \\P(E) &= \frac{n(E)}{n(S)} \\P(E) &= \frac{5}{23}\end{aligned}$$

To find probability not selecting a green ball, we have

$$\begin{aligned}P(E') &= 1 - P(E) \\P(E') &= 1 - \frac{5}{23} \\P(E') &= \frac{23 - 5}{23} \\P(E') &= \frac{18}{23}\end{aligned}$$

4. Three coins are tossed together. what is the probability of getting: (i) exactly three heads (ii) at least two tails (iii) not at least two heads (iv) not exactly two heads

When 3 coins are tossed, the sample space will be:

$$\begin{aligned}S &= \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\} \\n(S) &= 8\end{aligned}$$

(i) Let A be the event of getting exactly three heads.

$$\begin{aligned}A &= \{HHH\} \\n(A) &= 1 \\P(A) &= \frac{n(A)}{n(S)} \\P(A) &= \frac{1}{8}\end{aligned}$$

(ii) Let B be the event of getting at least two tails.

$$\begin{aligned}B &= \{HTT, THT, TTH, TTT\} \\n(B) &= 4 \\P(B) &= \frac{n(B)}{n(S)} \\P(B) &= \frac{4}{8}\end{aligned}$$

$$P(B) = \frac{1}{2}$$

(iii) Let C be the event of getting at least two heads.

$$C = \{HHT, HTH, THH, HHH\}$$

$$n(C) = 4$$

$$P(C) = \frac{n(C)}{n(S)}$$

$$P(C) = \frac{4}{8}$$

$$P(C) = \frac{1}{2}$$

To find probability of not getting at least two heads, we have

$$P(C') = 1 - P(C)$$

$$P(C') = 1 - \frac{1}{2}$$

$$P(C') = \frac{2-1}{2}$$

$$P(C') = \frac{1}{2}$$

(iv) Let D be the event of getting exactly two heads.

$$D = \{HHT, HTH, THH\}$$

$$n(D) = 3$$

$$P(D) = \frac{n(D)}{n(S)}$$

$$P(D) = \frac{3}{8}$$

To find probability of not getting exactly two heads, we have

$$P(D') = 1 - P(D)$$

$$P(D') = 1 - \frac{3}{8}$$

$$P(D') = \frac{8-3}{8}$$

$$P(D') = \frac{5}{8}$$

5. A card is drawn from a well shuffled pack of 52 playing cards. What will be the probability of getting: (i) king or jack of red colour (ii) not "2" of club and spade

$$\text{Total number of cards} = 52$$

$$n(S) = 52$$

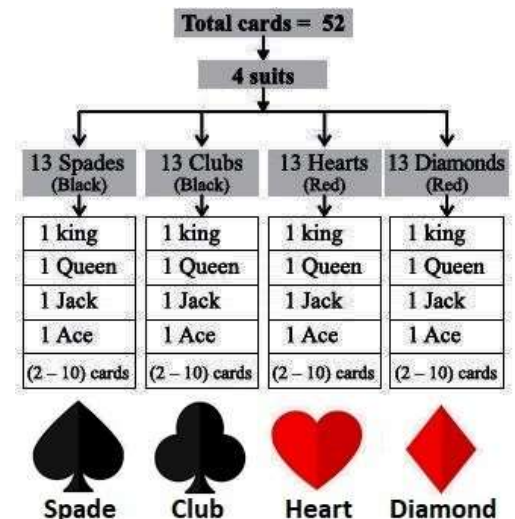
(i) Let A be the event of getting king or jack of red colour, so

$$n(A) = 4$$

$$P(A) = \frac{n(A)}{n(S)}$$

$$P(A) = \frac{4}{52}$$

$$P(A) = \frac{1}{13}$$



Note: First, identify the favourable cards:

- Red suits are Hearts and Diamonds.
- Each red suit has 1 King and 1 Jack.
- So, total red Kings = 2 (Hearts & Diamonds)
- Total red Jacks = 2 (Hearts & Diamonds)

(ii) Let B be the event of getting “2” of club and spade, so

$$n(B) = 2$$

$$P(B) = \frac{n(B)}{n(S)}$$

$$P(B) = \frac{2}{52}$$

$$P(B) = \frac{1}{26}$$

To find the probability of not getting “2” of club and spade, we have

$$P(B') = 1 - P(B)$$

$$P(B') = 1 - \frac{1}{26}$$

$$P(B') = \frac{26 - 1}{26}$$

$$P(B') = \frac{25}{26}$$

6. Six coins are tossed 600 times. The number of occurrence of tails are recorded and shown in the table given below:

No. of tails	0	1	2	3	4	5	6
Frequency	110	90	105	80	76	123	16

Find the relative frequency of given table.

No. of tails (x)	Frequency (f)	Relative Frequency ($\frac{f}{600}$)
0	110	$\frac{110}{600} = \frac{11}{60}$
1	90	$\frac{90}{600} = \frac{3}{20}$
2	105	$\frac{105}{600} = \frac{7}{40}$
3	80	$\frac{80}{600} = \frac{2}{15}$
4	76	$\frac{76}{600} = \frac{19}{150}$
5	123	$\frac{123}{600} = \frac{41}{200}$
6	16	$\frac{16}{600} = \frac{2}{75}$
Total	$\Sigma f = 600$	

7. From a lot containing 25 items, 8 items are defective. Find the relative frequency of non-defective items, also find the expected frequency of non-defective items.

$$\text{Total no. of items} = 25$$

$$n(S) = 25$$

Let A be the event of getting defective items

$$A = \text{No. of defective items}$$

$$n(A) = 8$$

$$P(A) = \frac{n(A)}{n(S)}$$

$$P(A) = \frac{8}{25}$$

To find the probability of getting non-defective items, we have

$$P(A') = 1 - P(A)$$

$$P(A') = 1 - \frac{8}{25}$$

$$P(A') = \frac{25 - 8}{25}$$

$$P(A') = \frac{17}{25}$$

Now by using formula

$$\text{Expected frequency of non defective} = \text{Total number of items} \times P(A')$$

$$E = N \times P(A')$$

$$E = 25 \times \frac{17}{25}$$

$$E = 17$$

OR

$$\text{Total items} = 25$$

$$\text{Defective items} = 8$$

$$\begin{aligned} \text{Non defective items} &= 25 - 8 \\ &= 17 \end{aligned}$$

$$\begin{aligned} \text{Relative Frequency} &= \frac{\text{Non - defective items}}{\text{Total items}} \\ &= \frac{17}{25} \end{aligned}$$

Now

$$\text{Expected Frequency} = \text{Relative Frequency} \times \text{Total items}$$

$$= \frac{17}{25} \times 25$$

$$= 17$$